

NGUYEN KHAC BAO

Intern AI/ML/Data Engineer

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CAREER OBJECTIVE

AI/ML/Data Engineer Intern candidate with hands-on experience building RAG chatbots, LLM integrations, and reliable data pipelines. Interested in applying Python, SQL, and cloud-based microservices to build AI agents for data analysis, reporting, and business automation.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

2022 – 2027

Bachelor of Information Technology

Relevant Coursework: Machine Learning, Deep Learning, Database Systems, Computer Vision.

TECHNICAL SKILLS

Programming Languages: Python, SQL (joins, CTEs, basic window functions), Bash

AI / LLMs: RAG pipelines, LlamaIndex ReAct, Embeddings/Vector DBs, Prompt Engineering, PEFT/LoRA, Gemini/OpenAI-compatible APIs

Data Engineering: Pandas, JSONL/CSV processing, ETL/ELT basics, data preprocessing, data quality checks

Backend / Tools: FastAPI, Docker, Qdrant, Git/GitHub, Linux, Streamlit

Basic Cloud & Messaging: Google Cloud Run (microservices), AWS EC2, Vast.ai (GPU), RabbitMQ

LANGUAGES

TOEIC: 610/990

2026

English: Technical reading proficiency

Able to read official documentation, research papers, model cards, API references, and engineering articles in English.

PROJECTS

Vietnamese Legal RAG Chatbot

2026

Personal Project | Technology: Python, FastAPI, Qdrant, LlamaIndex, Celery, Valkey, MariaDB, PEFT/LoRA, Gemini API | Source: [GitHub](#)

- Built an end-to-end Vietnamese legal assistant with FastAPI, Streamlit, asynchronous task handling, conversation storage, Qdrant vector retrieval, and OpenAI-compatible LLM endpoints.
- Designed an agent-style RAG workflow to route queries between legal retrieval, LlamaIndex ReAct tools, web-search handling, and chat fallback.
- Implemented a data import and indexing pipeline from JSONL legal QA records into Qdrant, with hybrid retrieval, query expansion, reranking, and legal-context prompting.
- Fine-tuned Llama-3.1-8B-Instruct with PEFT/LoRA on Vietnamese legal QA data using Vast.ai GPU; integrated Gemini/local LLM APIs for chatbot responses.
- Dockerized the multi-service system and added safe repository rules to avoid committing API keys, datasets, logs, and model artifacts.

FallGuard Labeling Tool – Dataset Assistant

2026

Personal Project | Technology: Python, Streamlit, OpenCV, Pandas, YOLO-Pose, CSV metadata | Source: [GitHub](#)

- Built a dataset assistant for fall-detection image/video data: raw import, frame extraction, metadata generation, auto-label initialization, human review, QA checks, split, and export.
- Designed CSV-based source-of-truth records for metadata, annotations, keypoints, and annotation events to keep data review and export reproducible.
- Integrated YOLO-Pose and a Streamlit review UI for label correction, uncertain/excluded samples, pose quality review, and manual keypoint editing.

Traffic Violation Detection & License Plate Recognition

2026

Personal Project | Technology: Python, OpenCV, YOLO, ByteTrack, HyperLPR3, NumPy, Shapely, JSON | Source: [GitHub](#)

- Built a fixed-camera traffic video pipeline for vehicle detection, tracking, traffic-light state estimation, stop-line violation logic, plate detection, OCR, and structured evidence generation.
- Fine-tuned a YOLO-based license plate detector on a compact CCPD2019 subset of 8,598 images, achieving 0.989 precision, 0.998 recall, and 0.994 mAP50 on validation.
- Implemented dataset QA checks for broken files, YOLO bbox validity, negative samples, OCR crop availability, duplicate review, low-light images, heavy blur, and small plates.

ACTIVITIES

AI Research Group – University Club

2025 – Present

Member

Discussed Machine Learning, RAG/LLM applications, Computer Vision, and AI system development.